

Erdős Problem 7: Odd Distinct Covering Systems

Theoretical Sieve Analysis, Prime-Power Obstruction, Candidate- L and Pruning Invariants

Investigation of Required Tasks 1, 3, 4, 7
 erdosproblems.com/7 — BBMST / Hough–Nielsen verification
 Compiled 2026-08-25

Abstract

We investigate Erdős Problem 7: does a finite distinct covering system with all moduli odd exist? Status is **OPEN** (verifiable). We verify against primary sources: Balister–Bollobás–Morris–Sahasrabudhe–Tiba [1] proves no odd *squarefree* distinct covering and, jointly with Hough–Nielsen [2], the restrictions $2 \mid d$ or $3 \mid d$ in any distinct covering and $9 \mid L$ or $15 \mid L$ for any hypothetical odd covering ($L = \text{lcm } d_i$). The folklore abundant condition $\sum_{d \mid L, d > 1} 1/d \geq 1 \iff \sigma(L) \geq 2L$ forces odd $L \geq 945$. We reconstruct the BBMST distortion/martingale sieve in its actual notation, define $c_N(x) = c_0(x) \prod_k U_{p_k}(x)$ with $U_{p_k}(x) = 1 + x/((1 - \delta_k)s_k)$, $s_k = p_k - 1$, $c_0(1) \approx 0.098$, $c_N(1) \approx 0.612 < 1$. Extending to $d = \prod p^{a_p}$ with some $a_p > 1$, we isolate the *first* obstruction: fibre coupling $\alpha_i(x) = \sum_{j=1}^{\gamma_i} \sum_m p_i^{-j} \mathbf{1}[x \equiv a_{mp_i^j} \pmod{m}]$, non-factorisation of $U^{(\gamma)}$, non-transfer of the LP optimum c_0 , and δ -exponent dependence. Naive monotonicity $p^e - 1 \geq p - 1 \Rightarrow \prod(1 + x/((1 - \delta)s^{\text{pow}})) \leq \prod(1 + x/((1 - \delta)s^{\text{SF}}))$ is structurally false even after inserting c_0 ; second-moment prefactor is $(\sum_j p^{-j})^2 = (1 - p^{-\gamma})^2/(p - 1)^2$ which *increases* with γ . For $L \in \{945, 1575, 2205, 2835, \dots\}$ we compute exact factorisations, divisor counts, reciprocal sums, and prove no odd $L < 945$ reaches the density threshold. Minimal feasible supports have $k \geq 10$ moduli and $\sum 1/d$ exceeds 1 by only 0.016–0.049. We give exact SAT/ILP encodings ($V = \sum_{d \mid L} d = 1919$ –5807 variables, L covering clauses, AMO constraints) and derive provably safe pruning invariants (reciprocal sum, density deficit, CRT interference, LCM anchoring, translation symmetry) distinguished from heuristic ordering. All numerics are verified by `ANSWER/verify_solution.py` with residual 0. Ten-point self-audit is performed. The barrier for general odd moduli is technical, not proved intrinsic: conditional Lemma C (certified LP $c_0^{\text{pow}}(1) \leq c_0^{\text{SF}}(1)$) would bridge the gap, but is not proved and must be recomputed over 27 vs 16 numbers ≤ 73 .

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1. Problem Statement

Definition 1.1 (Covering system). A finite family $\mathcal{C} = \{a_i \bmod m_i\}_{i=1}^k$ with $m_i > 1$, $a_i \in \mathbb{Z}$ is a *covering system* if

$$\bigcup_{i=1}^k (a_i + m_i \mathbb{Z}) = \mathbb{Z},$$

i.e. every integer satisfies at least one congruence. It is *distinct* if m_i are pairwise distinct, and *odd* if $2 \nmid m_i$ for all i .

Put $L := \text{lcm}(m_1, \dots, m_k)$. Then R_i periodic mod L and $\mathcal{C}$ covers $\mathbb{Z}$ iff it covers $[0, L - 1]$ iff $\forall n \in \mathbb{Z}/L\mathbb{Z} \exists i : n \equiv a_i \pmod{m_i}$. Each $m_i \mid L$.

Erdős Problem 7: Is there a distinct covering system all of whose moduli are odd? Erdős–Selfridge conjecture: no. Asker: Erdős–Selfridge (sometimes Schinzel). Status on <https://www.erdosproblems.com/7>: **VERIFIABLE Open**, \$25. Also asks squarefree variant. Formalised in `google-deepmind/formal-conjectures` file `FormalConjectures/ErdosProblems/7.lean` as `answer(sorry)`.

Infinite odd distinct almost-covering exists (e.g. $\{0 \bmod p_{k+1}, 0 \bmod p_{k+2}, \dots\}$ for odd primes p_j); finite almost-cover $\Rightarrow$ cover by periodicity mod L . So finiteness essential.

2. Verified Known Results

We distinguish **proved** vs **conjectural** against primary literature, not secondary summaries. Sources fetched 2026-08-25: problem page, `/latex/7`, `/history/7`, `/forum/thread/7` (21 comments), bibliography entries, database `data/problems.yaml#7`.

Theorem 2.1 (BBMST22, Invent. Math. 230 (2022), 377–414, Thm 1.1; arXiv:1811.03547 Thm 1.3 + companion arXiv:1901.11465). *There is no distinct covering system all of whose moduli are odd and squarefree.*

Quotation from 1811.03547 p. 8: “Unfortunately, our method does not seem to be strong enough to resolve the Erdős–Selfridge problem (see the discussion in Section 6). However, it does allow us to make some further progress. . . we can prove that no such covering system exists under the additional constraint that the moduli are square-free. Since this application. . . requires several additional (somewhat technical) ideas, we will give the details elsewhere [1].” Reference [1] is 1901.11465.

Theorem 2.2 (Hough–Nielsen, Duke Math. J. 168 (2019), 3261–3295, arXiv:1703.02133; simplified proof in BBMST22). *In any distinct covering system (no parity assumption), at least one modulus is divisible by 2 or 3.*

Theorem 2.3 (BBMST22, Thm 1.4 / Cor; strengthening of HoNi for odd case). *If a distinct odd covering system exists and $L = \text{lcm}(m_i)$, then $9 \mid L$ or $15 \mid L$.*

Proposition 2.4 (Folklore, Rafik Zenodo 18360978, comment 3717). *Let $\mathcal{C}$ be distinct covering with $L = \text{lcm}(m_i)$. Then*

$$\sum_i \frac{1}{m_i} \geq 1, \quad \sum_{\substack{d \mid L \\ d > 1}} \frac{1}{d} \geq \sum_i \frac{1}{m_i} \geq 1 \iff \frac{\sigma(L)}{L} \geq 2,$$

so L is abundant. If all m_i odd then L is odd abundant, hence $L \geq 945 = 3^3 \cdot 5 \cdot 7$, the smallest odd abundant (OEIS A005231), $\sigma(945) = 1920$.

Proof. Each $a_i \bmod m_i$ contributes L/m_i residues mod L . To cover $[0, L-1]$ need $\sum L/m_i \geq L$. Since $\{m_i\} \subseteq \{d \mid L, d > 1\}$, the sum over all divisors dominates. The identity $\sigma(L) = \sum_{d \mid L} d = L \sum_{d \mid L} 1/d$ gives equivalence. Minimality of 945 is by direct divisor-sum enumeration; see `verify_solution.py` exhaustive check: no odd $L < 945$ has $\sigma(L) \geq 2L$ or $\sum_{d \mid L, d > 1} 1/d \geq 1$. $\square$

Conjectural: Full Erdős–Selfridge “no odd distinct covering at all (allowing prime powers)” remains **open**; no counterexample exhibited, no proof published. Two 2026 Lean attempts (“spicylemonade/erdos-007” two axioms, “axxen95” three axioms via sieve monotonicity) are invalid: see forum #6298 natso26 refutation – `sieveProd` = $\prod (1 + x/s)$ with $s > 0$ is always > 1 at $x = 1$, so axiom `sieveProd < 1` contradictory; missing baseline LP value $c_0 \approx 0.098$. Author #6302 admits omission; moderator #6316 blocks further axiomatic discussion without complete sorry/axiom-free formalisation.

Selfridge reduction via [3] (odd covering would follow from covering with $n_i \nmid n_j$) is **proved irrelevant**: antecedent does not exist by [?] (see problem 586).

3 3. Exact Definitions

We fix notation as in BBMST 1811.03547 §§2–3.

Let $\mathcal{A} = \{A_d : d \in D\}$, $A_d = a_d + d\mathbb{Z}$, D finite distinct odd, $Q := \text{lcm}(D) = \prod_{j=1}^n p_j^{\gamma_j}$, $\gamma_j \geq 1$, p_j odd primes ordered (typically increasing). Define

$$\begin{aligned} Q_i &:= \prod_{j=1}^i p_j^{\gamma_j}, & Q_0 &:= 1, \\ D_i &:= \{d \in D : d \mid Q_i\}, & \mathcal{A}_i &:= \{A_d : d \in D_i\}, \\ R_i &:= \mathbb{Z} \setminus \bigcup_{d \in D_i} A_d, & R_0 &:= \mathbb{Z}, R_n = R, \\ N_i &:= D_i \setminus D_{i-1} \quad (\text{new moduli at step } i), \\ B_i &:= \bigcup_{d \in N_i} A_d, & \text{so } R_i &= R_{i-1} \setminus B_i. \end{aligned} \tag{1811.03547 (3)}$$

Each R_i, B_i is Q_i -periodic, identified with subset of $\mathbb{Z}_{Q_i}$.

Divisor notation: $\tau(L)$ = number of divisors, $\sigma(L) = \sum_{d \mid L} d$, $D(L) = \{d \mid L : d > 1\}$, $V(L) = \sum_{d \in D(L)} d$ = number of SAT variables $x_{d,a}$.

4 4. Reconstruction of the Relevant BBMST Machinery

We reconstruct the distortion process in *actual* notation, not an invented analogue. All equation numbers refer to arXiv:1811.03547 v1.

4.1 Prime exposure and fibres

Factor $\mathbb{Z}_{Q_i} \cong \mathbb{Z}_{Q_{i-1}} \times \mathbb{Z}_{p_i^{\gamma_i}}$ (CRT, coprime). For $x \in \mathbb{Z}_{Q_{i-1}}$,

$$F(x) := \{(x, y) : y \in \mathbb{Z}_{p_i^{\gamma_i}}\} \subseteq \mathbb{Z}_{Q_i}, \quad |F(x)| = p_i^{\gamma_i}.$$

4.2 Fibre proportion $\alpha_i(x)$

Uniform count proportion

$$\alpha_i(x) := \frac{|F(x) \cap B_i|}{|F(x)|}.$$

More precisely under distorted measure, with $d = mp_i^j \in N_i$, $1 \leq j \leq \gamma_i$, $m \mid Q_{i-1}$ unique,

$$\alpha_i(x) \leq \sum_{d=mp_i^j \in N_i} p_i^{-j} \mathbf{1}[x \equiv a_d \pmod{m}] \quad (34-35)$$

where the condition is Q_{i-1} -measurable. Inclusion–exclusion overlap gives error $E_i(x) \geq 0$ with α_i equal to the sum minus overlaps; squarefree case $\gamma_i \equiv 1$ reduces to single denominator p_i^{-1} and $N_i = \{mp_i : m \mid Q_{i-1}\}$.

No scalar $p_i^{\gamma_i} - 1$ appears as block size; effective weight is $\sum_j p_i^{-j} = (1 - p_i^{-\gamma_i})/(p_i - 1)$.

4.3 Distorted measures $\mathbb{P}_i$ and parameters δ_i

Fix $\delta_i \in [0, 1/2]$. $\mathbb{P}_0$ uniform on $\mathbb{Z}_Q$. Inductively $\mathbb{P}_i$ is the unique probability on $\mathbb{Z}_{Q_i}$ coinciding with $\mathbb{P}_{i-1}$ on Q_{i-1} -measurable sets and redistributing mass fibrewise to cap B_i at δ_i : for $x \in \mathbb{Z}_{Q_{i-1}}$,

- if $\alpha_i(x) \leq \delta_i$, points of $F(x) \cap B_i$ down-weighted by $(\alpha_i - \delta_i)/(\alpha_i(1 - \delta_i))$ and points of $F(x) \setminus B_i$ up-weighted by $1/(1 - \delta_i)$;
- if $\alpha_i(x) > \delta_i$, no distortion on $F(x)$.

Consequence (Lemma 3.3, (16)):

$$\mathbb{P}_i(B_i) \leq \min\left\{\mathbb{E}_{i-1}[\alpha_i], \frac{\mathbb{E}_{i-1}[\alpha_i^2]}{4\delta_i(1 - \delta_i)}\right\},$$

where $\mathbb{E}_{i-1}$ is expectation under $\mathbb{P}_{i-1}$ on $\mathbb{Z}_{Q_{i-1}}$. Pointwise distortion $\log(\mathbb{P}_j(x)/\mathbb{P}_{j-1}(x)) \leq 2\alpha_j(x)$ (24); Lemma 3.5 controls $\mathbb{E}_i[\Delta_i] \leq 2 \sum_{j \leq i} \mathbb{E}_{j-1}[\alpha_j]$, $\Delta_i(x) = \sum_{j \leq i} \log(\mathbb{P}_j(x)/\mathbb{P}_{j-1}(x))$.

4.4 Moment functionals

$$M_i^{(1)} := \mathbb{E}_{i-1}[\alpha_i], \quad M_i^{(2)} := \mathbb{E}_{i-1}[\alpha_i^2]. \quad (13)$$

Theorem 3.2 (general k th moment, (36)):

$$\mathbb{E}_{i-1}[\alpha_i^k] \leq \sum_{d_1=m_1 p_i^{j_1} \in N_i} \cdots \sum_{d_k=m_k p_i^{j_k} \in N_i} \frac{1}{p_i^{j_1+\cdots+j_k}} \mathbb{P}_{i-1}(x \equiv a_{d_t} \pmod{m_t} \forall t).$$

Bounding uses $\nu(m) := \max_a \mathbb{P}_{i-1}(x \equiv a \pmod{m}) \cdot m$ and inductively $\mathbb{P}_{i-1}(x \equiv a \pmod{m}) \leq \nu(m)/m$ (38), with

$$\sum_{m \mid Q_{i-1}} \frac{\nu(m)}{m} \leq \prod_{j < i} \left(1 + \frac{1}{(1 - \delta_j)(p_j - 1)}\right), \quad M_i^{(2)} \leq \frac{1}{(p_i - 1)^2} \prod_{j < i} \left(1 + \frac{3p_j - 1}{(1 - \delta_j)(p_j - 1)^2}\right). \quad (47, \text{squarefree form})$$

For general γ_i the prefactor is *not* $1/(p_i^{\gamma_i} - 1)^2$; summing $\sum_{j_1, j_2 \geq 1} p_i^{-(j_1+j_2)} = 1/(p_i - 1)^2 \cdot (1 - p_i^{-\gamma_i})^2$ leaves denominator $(p_i - 1)^2$ and adds combinatorial multiplicity (more terms in N_i).

4.5 Accumulated operator $c_N(x) = c_0(x) \prod U_{p_k}(x)$

Package Sections 5–6 and 1901.11465 §3: define numerical sieve

$$f_k := \text{distortion-corrected uncovered parameter after } p_1, \dots, p_k,$$

with recursion (Lemma 6.2, (23)):

$$f_i \leq \left(1 + \frac{3p_i - 1}{(1 - \delta_i)(p_i - 1)^2}\right) \left(1 - \frac{f_{i-1}}{4\delta_i(1 - \delta_i)(p_i - 1)^2}\right)^{-1} f_{i-1},$$

when $4b_i f_{i-1} < 1$, $b_i := M_i^{(2)}/f_{i-1}$, and optimal $\delta_i = (1 + a_i)/(1 + \sqrt{1 + a_i(1 + a_i)/(b_i f_{i-1})})$, $a_i := (3p_i - 1)/((1 - \delta_i)(p_i - 1)^2)$ implicit (25).

Writing as multiplicative update,

$$\boxed{c_N(x) = c_0(x) \cdot \prod_{k=1}^N U_{p_k}(x), \quad U_{p_k}(x) := 1 + \frac{x}{(1 - \delta_k)s_k}, \quad s_k := p_k - 1 \text{ (squarefree)}} \quad (\star)$$

where $c_0(x)$ is baseline LP value and each U comes from second-moment bound (47). For general γ_k ,

$$U_{p_k}^{(\gamma_k)}(x) = 1 + \frac{x}{1 - \delta_k} \omega_k^{(\gamma_k)},$$

$\omega_k^{(\gamma_k)}$ encodes $\sum_j p_k^{-j}$ and ν -sums, not $1/(p_k^{\gamma_k} - 1)$.

4.6 Baseline LP value $c_0(x)$

For small primes (first 5 primes $p \leq 13$ or full $p \leq 73$, $k_0 = 21$) BBMST do *not* use moment bound; they enumerate partition of $\mathbb{Z}_{Q_0}$, $Q_0 = \prod_{j \leq k_0} p_j$ (squarefree $\prod p_j$ atoms; with prime powers $\prod p_j^{\gamma_j}$ atoms exponentially more) and solve LP to minimise maximal fibre density of uncovered set under admissible residue assignments. Numerically (1901.11465 §5, $N = 500$):

$$\boxed{c_0(1) \approx 0.097\text{--}0.098}, \quad \prod_k U_{p_k}(1) \approx 6.25, \quad c_N(1) = c_0(1) \prod U_{p_k}(1) \approx 0.612 < 1.$$

Because $c_0 < 1$ and each $U > 1$, product staying < 1 is non-trivial. The false Lean axiom `sieveProd` = $\prod(1 + x/s) < 1$ omitted c_0 ; indeed $\prod(1 + x/s) > 1$ for any $x = 1$, $s > 0$, $U > 1$.

Prime-power $c_0^{(\gamma)}(x)$ optimises over finer partition with moduli $p_j^{\gamma_j}$; feasible region differs: more variables (each mp_j^j) and constraints. No theorem gives $c_0^{(\gamma)} \leq c_0^{\text{SF}}$; allowing 9, 25, 27, 45, 49, ... enlarges adversary's options, likely *increasing* LP optimum.

4.7 Distortion dependencies

$$\delta_i^{(\gamma)} = \Phi(p_i, \gamma_i, f_{i-1}^{(\gamma)}, \{\delta_j^{(\gamma)}\}_{j < i})$$

for explicit Φ from (25) involving a_i, b_i which depend on γ_i via $M_i^{(2)}$. So δ not exponent-independent, contrary to post #6289 claim.

5 Prime-Power Extension Analysis

Let $d = \prod_p p^{a_p}$, some $a_p > 1$, $\gamma_i = a_{p_i}$.

For each of 7 required points we state squarefree step, prime-power analogue, inequality, usage of squarefreeness, martingale survival, quantified extra distortion, summability, and first obstruction.

1. Exact squarefree step being generalized

Squarefree (SF): $N_i = \{mp_i : m \mid Q_{i-1}\}$, $\alpha_i^{\text{SF}}(x) \leq \sum_{mp_i \in N_i} p_i^{-1} \mathbf{1}[x \equiv a_{mp_i} \pmod{m}]$, moment bound $M_i^{(2)} \leq (p_i - 1)^{-2} \prod_{j < i} (1 + (3p_j - 1)/((1 - \delta_j)(p_j - 1)^2))$.

2. Prime-power statement

PP: $N_i = \{mp_i^j : 1 \leq j \leq \gamma_i, m \mid Q_{i-1}\}$,

$$\alpha_i^{(\gamma)}(x) = \sum_{j=1}^{\gamma_i} \sum_{m: mp_i^j \in N_i} p_i^{-j} \mathbf{1}[x \equiv a_{mp_i^j} \pmod{m}] + E_i(x),$$

E_i inclusion-exclusion overlaps $0 \leq E_i \leq \sum_{d \neq d'} \text{cross terms}$.

3. Replacement inequality

Naive proposed replacement:

$$U_{p_i}^{(\gamma)}(x) \stackrel{?}{=} 1 + \frac{x}{(1 - \delta_i)(p_i^{\gamma_i} - 1)} \quad (\text{false}).$$

Correct replacement from (36)–(47) summed over j_1, j_2 :

$$M_i^{(2, \gamma)} \leq \left(\sum_{j=1}^{\gamma_i} p_i^{-j} \right)^2 \sum_{m_1, m_2 \mid Q_{i-1}} \frac{\nu(\text{lcm})}{\text{lcm}} = \frac{(1 - p_i^{-\gamma_i})^2}{(p_i - 1)^2} \prod_{j < i} \left(1 + \frac{3p_j - 1}{(1 - \delta_j)(p_j - 1)^2} \right)^{(\gamma_{< i})}.$$

Hence effective factor $1 + x\eta_i^{(\gamma)}/(1 - \delta_i)$ with $\eta_i^{(\gamma)} \asymp (1 - p_i^{-\gamma_i})^2/(p_i - 1)^2$ times history product.

Naive inequality $1 + x/((1 - \delta)(p^\gamma - 1)) \leq 1 + x/((1 - \delta)(p - 1))$ is *true pointwise* but *irrelevant*: true factor uses $(1 - p^{-\gamma})^2/(p - 1)^2$ which *increases* with γ (since $(1 - p^{-\gamma}) \uparrow 1$), opposite direction to $1/(p^\gamma - 1)$ which *decreases*.

$$\frac{(1 - p^{-\gamma})^2}{(p - 1)^2} \xrightarrow{\gamma \rightarrow \infty} \frac{1}{(p - 1)^2}, \quad \frac{1}{p^\gamma - 1} \xrightarrow{\gamma \rightarrow \infty} 0.$$

So second-moment coupling gets *larger* (worse for sieve), not smaller.

4. Where squarefreeness used

- Fibre size $|F(x)| = p_i$ vs $p_i^{\gamma_i}$: bijection $\mathbb{Z}_{Q_i} \cong \mathbb{Z}_{Q_{i-1}} \times \mathbb{Z}_{p_i}$ used to define uniform weight p_i^{-1} ; with $\gamma_i > 1$ fibre splits into $p_i^{\gamma_i}$ points but condition $x \equiv a_d \pmod{m}$ does not depend on $y \in \mathbb{Z}_{p_i^{\gamma_i}}$ alone, coupling across exponents.
- Enumeration of N_i : SF has at most $\tau(Q_{i-1})$ new moduli; PP has up to $\gamma_i \tau(Q_{i-1})$ (factor γ_i more), entering moment sums.
- LP baseline c_0 : SF atoms $\prod_{j \leq k_0} p_j$ ($2.0\text{--}7.5 \times 10^9$ for $p \leq 73$ already large but feasible via symmetry); PP atoms $\prod p_j^{\gamma_j}$ exponential in $\sum \gamma_j$, intractable without new symmetry reduction.
- ν -history product $\prod_{j < i} (1 + O(1/p_j^2))$ derived assuming $p_j - 1$ scalar block; PP history uses $\sum_j p_j^{-j}$ weights, changing constants $3p - 1$ to more complicated $\Theta_i(s, t)$ recursion (84).

5. Does martingale/independence survive?

Distorted measure construction survives *formally* (still defines $\mathbb{P}_i$ fibrewise). What fails is *independence estimate* in Lemma 3.7 / (36): terms with distinct exponents $j_1 \neq j_2$ for same m are *not* independent and inclusion-exclusion E_i not controlled by pairwise lcm bound uniformly in γ . The “good fibre” second-moment bound $M_i^{(2)}/(4\delta(1 - \delta))$ assumes α_i is sum of *disjoint* indicators with common denominator p_i ; with mixed denominators p_i^{-j} , variance inflates by γ_i factor before normalisation. Martingale L^2 bound $\mathbb{E}_i[\Delta_i] \leq 2 \sum \mathbb{E}[\alpha]$ still holds but with larger $\mathbb{E}[\alpha]$ due to $\sum_j p^{-j}$ vs p^{-1} , so accumulated distortion $\sum \delta_i/(1 - \delta_i)$ grows faster.

6. Quantified additional error/distortion

Early step $p_i = 3, \gamma = 3$: SF weight $p^{-1} = 1/3 \approx 0.333$; PP weight $\sum_{j=1}^3 p^{-j} = 1/3 + 1/9 + 1/27 = 13/27 \approx 0.481$ (+44%). Squared for $M^{(2)}$: $(1/3)^2 = 0.111$ vs $(13/27)^2 \approx 0.232$ (2.09× larger). For $p = 5, \gamma = 2$: 0.2 vs 0.24 (+20%), squared 0.04 vs 0.0576 (1.44×). For $p = 7, \gamma = 2$: 0.1429 vs 0.1633, squared 0.0204 vs 0.0267 (1.31×). History product inherits similar inflation compounding over i .

Effective s -monotonicity claimed $s^{\text{pow}} \geq s^{\text{SF}}$ would predict *decrease* of $M^{(2)}$ by factor $(p-1)^2/(p^\gamma-1)^2 \approx p^{-2(\gamma-1)}$ (e.g. $p = 3, \gamma = 3$: $1/26^2$ vs $1/2^2 = 1/676$ reduction 169×), opposite to true 2.09× *increase*. Error is order-of-magnitude sign reversal.

7. Summability / accumulated error

Need $c_N^{(\gamma)}(1) = c_0^{(\gamma)}(1) \prod_{k \geq k_0} (1 + \eta_k^{(\gamma)} / (1 - \delta_k^{(\gamma)})) < 1$. Even if tail factors individually $\approx 1 + O(1/p_k^2)$, sum $\sum_k \eta_k^{(\gamma)} / (1 - \delta_k)$ must stay $\ll \log(1/c_0^{(\gamma)})$. Since $\eta_k^{(\gamma)} \geq 1/(p_k-1)^2 \cdot (1+p_k^{-1})^2$ *larger* than SF, and $c_0^{(\gamma)} \geq c_0^{\text{SF}}$ plausibly (LP feasible region larger), accumulated error is *strictly larger* than 0.612 baseline. Whether it stays < 1 is *unproved* and cannot be deduced by monotonicity; requires recomputing $c_0^{(\gamma)}$ and re-optimising $\delta^{(\gamma)}$ over tail $p \leq 500$. No summability collapse to < 1 follows without new computation.

First obstruction isolated

FIRST obstruction = fibre coupling across exponents j in $\alpha_i(x)$ and consequent $M_i^{(2)}$ prefactor $(\sum_j p_i^{-j})^2$ in

Naive block-size monotonicity $p^\gamma - 1$ mistakes the fibre cardinality for the moment weight; weight is $\sum p^{-j}$, which grows with γ , making second moment larger, LP feasible region larger, and $c_0^{(\gamma)}$ potentially larger. Correct comparison keeps denominator $(p-1)^2$ fixed and multiplies by $(1-p^{-\gamma})^2 > 1$ over SF. Hence product $\prod (1 + x/((1-\delta)(p^\gamma-1)))$ is *not* the true BBMST operator; true operator's factor is $1 + x\eta^{(\gamma)}/(1-\delta)$ with $\eta^{(\gamma)}$ *increasing* in γ . This breaks factorwise monotonicity at the *first* fibre definition (34)–(35) before any LP or δ optimisation.

Consequences: $c_0^{(\gamma)}$ not monotone, $\delta^{(\gamma)}$ exponent-dependent via b_i , and tail product monotonicity claim (still true for hypothetical scalar s) does not apply to actual PP operator which has different η weights and ν -history. Overcoming requires re-solving LP over 27 vs 16 numbers ≤ 73 and certifying new δ sequence remains feasible – the *only* unproved component (Conditional Lemma C).

Barrier classification: **technical, not proved intrinsic**. No evidence distortion method fundamentally incapable; tail heuristics suggest extension plausible (p^γ helps covering intuition but sieve second moment worsens). BBMST deliberately stopped at SF because initial segment already delicate; not trivial corollary.

6. Computational Formulation

Fix odd abundant L with $9 \mid L$ or $15 \mid L$. Let $D(L) = \{d \mid L : d > 1\}$ (size $t = |D|$). Every modulus of a covering with $\text{lcm} = L$ lies in $D(L)$. We formulate exact finite decision.

6.1 SAT/CNF encoding

Variables for each $d \in D, a \in [0, d-1]$:

$x_{d,a} \in \{0, 1\}, x_{d,a} = 1 \iff$ modulus d used with residue $a, \quad y_d \in \{0, 1\}, y_d = 1 \iff d$ used.

We also write y_d for y_d and $x_{d,a}$ for $x_{d,a}$ in code. Total $V_{\text{SAT}} = \sum_{d \in D} d$ (+ $|D|$ if y included): 1919–5807 for L in $\mathcal{L}$.

Exact SAT/ILP variables and constraints (required spec):

- **Candidate moduli:** $D(L) = \{d \mid L : d > 1, d \text{ odd}\}$ ($t = |D| = 15\text{--}19$). Every modulus of a covering with $\text{lcm} = L$ lies in $D(L)$.
- **Variables:** $x_{d,a} \in \{0, 1\}$ for $d \in D(L)$, $a \in [0, d-1]$; $y_d \in \{0, 1\}$ with linking $y_d = 1 \iff \exists a : x_{d,a} = 1$.
- **Oddness:** by construction $D(L)$ contains only odd divisors; enforced as $\forall d \text{ even} : y_d = 0$, $x_{d,a} = 0$ (no variable created). For odd L this is $D(L) = \{d \mid L : d > 1\}$ automatically odd.
- **Distinctness:** $\sum_{a=0}^{d-1} x_{d,a} = y_d \leq 1$ for each d (at most one residue per modulus: AMO). Satisfies $\sum_a x_{d,a} = y_d$ and $\sum_a x_{d,a} \leq 1$.
- **Congruence-class choice:** selection of a per used d is exactly $x_{d,a} = 1 \iff$ progression $a \bmod d$ included.
- **Coverage:** for every residue $r \in [0, L-1]$,

$$\boxed{\sum_{d \in D(L)} x_{d, r \bmod d} \geq 1 \quad \forall r \in [0, L-1]}, \quad \text{CNF} \quad \bigvee_{d \in D} x_{d, r \bmod d}.$$

L clauses width t , literals $L \cdot t$.

- **LCM consistency:** $\forall d (y_d = 1 \Rightarrow d \mid L)$ (by $d \in D(L)$) and anchoring $\forall p^e \parallel L : \sum_{d: p^e \mid d} y_d \geq 1$ (a multiple of each prime power divides some modulus), plus optional $\text{lcm}(\{d : y_d = 1\}) = L$.
- **Optional minimality:** $\sum_d y_d$ minimal or $\sum y_d/d$ cut.

CNF details:

- (C1) AMO per modulus: $\sum_a x_{d,a} \leq 1$, linking $\sum_a x_{d,a} = y_d$. CNF: pairwise $\bigwedge_{a < b} (\neg x_{d,a} \vee \neg x_{d,b})$ – $\sum \binom{d}{2}$ binary clauses (532k for 945, 1.48M for 1575, 2.90M for 2205, 4.79M for 2835) or sequential ladder $O(d)$ auxiliaries ($\approx 3 \sum d$ clauses: 5757–17421).
- (C2) Covering: as boxed above, L clauses $\sum_d x_{d, r \bmod d} \geq 1$.
- (C3) Oddness filter: $D(L)$ odd-only; even- d variables eliminated (constraint $y_d = 0$).
- (C4) LCM anchoring: $\forall p^e \parallel L \bigvee_{d: p^e \mid d} y_d$ (unit clause on y).
- (C5) Symmetry breaking (see §8): translation C_L invariance $x_{d,a} \mapsto x_{d, (a+t) \bmod d}$; fix $x_{3,0} = 1$.

6.2 ILP 0–1 formulation

$$\begin{array}{ll}
\sum_{a=0}^{d-1} x_{d,a} = y_d \leq 1 & \forall d \in D(L) \quad (\text{distinctness } \sum_a x_{d,a} = y_d \leq 1) \\
x_{d,a} \leq y_d & \forall d, a \\
x_{d,a}, y_d \in \{0, 1\} & (\text{variables } x_{d,a} \in \{0, 1\}, y_d \in \{0, 1\}) \\
y_d = 0 & \forall d \notin D(L) \text{ or } 2 \mid d \quad (\text{oddness}) \\
\sum_{d \in D(L)} x_{d,r \bmod d} \geq 1 & \forall r \in [0, L-1] \quad (\text{coverage}) \\
\sum_{d: p^e \parallel d} y_d \geq 1 & \forall p^e \parallel L \quad (\text{LCM consistency}) \\
x_{3,0} = 1 & (\text{translation fix}) \\
\sum_d y_d/d \geq 1 & (\text{reciprocal cut})
\end{array}$$

Dimensions: vars $V + |D|$, constraints $|D| + L$, nonzeros $\approx V + \sum_d L/d$ sparse (16k–60k). Matrix entry $x_{d,a}$ appears in 1 linking equality and L/d covering rows.

SMT variant: integer $a_d \in [-1, d-1]$ with -1 unused and clause $\forall n \bigvee_d (a_d \geq 0 \wedge n \equiv a_d \pmod{d})$, bit-blasted to SAT.

Correctness: assignment satisfying (C1)–(C3) and AMO yields $M = \{d : y_d = 1\}$, a_d unique with $x_{d,a_d} = 1$, distinct odd, covering $\mathbb{Z}/L\mathbb{Z}$ hence $\mathbb{Z}$; conversely any covering induces such assignment (after possibly fixing t translation). Finite decision per L .

7 7. Candidate- L Analysis

Necessary conditions: L odd abundant ($\sigma(L) \geq 2L$) and $9 \mid L$ or $15 \mid L$ (Thm 2.3). Smallest odd abundant is 945.

We computed exactly via SymPy (see `verify_solution.py`):

L	factorisation	$\tau(L)$	$ D $	$V = \sum D$	$\sum_{d \mid L, d > 1} 1/d$	$\sigma(L)/L$	$9 \mid L$ or $15 \mid L$
945	$3^3 \cdot 5 \cdot 7$	16	15	1919	$65/63 \approx 1.031746$	$128/63 \approx 2.031746$	yes / yes
1575	$3^2 \cdot 5^2 \cdot 7$	18	17	3223	$1649/1575 \approx 1.046984$	$3224/1575 \approx 2.046984$	yes / yes
2205	$3^2 \cdot 5 \cdot 7^2$	18	17	4445	$249/245 \approx 1.016327$	$494/245 \approx 2.016327$	yes / yes
2835	$3^4 \cdot 5 \cdot 7$	20	19	5807	$991/945 \approx 1.048677$	$1936/945 \approx 2.048677$	yes / yes
3465	$3^2 \cdot 5 \cdot 7 \cdot 11$	24	23	7487	$447/385 \approx 1.161039$	$832/385 \approx 2.161039$	yes / yes

Full divisor sets:

$$\begin{aligned}
D(945) &= \{3, 5, 7, 9, 15, 21, 27, 35, 45, 63, 105, 135, 189, 315, 945\} \\
D(1575) &= \{3, 5, 7, 9, 15, 21, 25, 35, 45, 63, 75, 105, 175, 225, 315, 525, 1575\} \\
D(2205) &= \{3, 5, 7, 9, 15, 21, 35, 45, 49, 63, 105, 147, 245, 315, 441, 735, 2205\} \\
D(2835) &= \{3, 5, 7, 9, 15, 21, 27, 35, 45, 63, 81, 105, 135, 189, 315, 405, 567, 945, 2835\}
\end{aligned}$$

Odd divisors = all (since L odd). Reciprocal sums 1.016–1.049 barely above 1; headroom ≤ 0.049 . Exhaustive scan confirms **no odd** $L < 945$ **has** $\sum_{d \mid L, d > 1} 1/d \geq 1$ (verified brute-force $3 \leq L < 945$ odd, 0 successes). Hence $L \geq 945$ sharp and necessary.

Not exhaustive without proof: Set $\mathcal{L} = \{945, 1575, 2205, 2835\}$ comprises four smallest odd abundant with $9 \mid L$ or $15 \mid L$ (checked up to 4000; next is 3465). We do **not** assume these

are the only candidates: any odd covering could have larger L (e.g. 3465, 4095, 4725, ...) and larger L does not logically contain a smaller L as divisor (e.g. 4725 = $3^3 5^2 7$ not multiple of 1575). So search over $\mathcal{L}$ yields conditional certificates “no covering with lcm in $\mathcal{L}$ ” but not global non-existence; sieve induction needed for unbounded L . Our reduction treats each L as maximal divisor universe: $M \subseteq D(L)$, so fixing L subsumes all M with $\text{lcm} \mid L$.

Combinatorial bounds: $|D| = 15\text{--}19$, naive residue assignments $\prod_{d \in D} d$ have $\log_{10} = 23.80, 28.78, 30.09, 34.53$ respectively (6.3×10^{23} to 3.4×10^{34}); after fixing $a_3 = 0$ divide by 3. Greedy minimal feasible support (10 smallest divisors reaching $\sum \geq 1$ for 945: $\{3, 5, 7, 9, 15, 21, 27, 35, 45, 63\}$, sum ≈ 1.005) still has $\prod \approx 8.0 \times 10^{11}$. Larger L grows monotonically in V and τ .

8 Exact Pruning Rules

We derive invariants; each flagged **SAFE** (sound, never discards feasible solution) vs **HEURISTIC** (ordering, not pruning).

1. **Reciprocal-density upper bound (SAFE)**. For partial $M_{\text{fixed}} = \{d : y_d = 1\}$, $M_{\text{open}} = \{d \text{ unassigned}\}$,

$$U := \sum_{M_{\text{fixed}}} 1/d + \sum_{M_{\text{open}}} 1/d < 1 \Rightarrow \text{prune.}$$

Proof: any extension $M \supseteq M_{\text{fixed}}$, $M \subseteq M_{\text{fixed}} \cup M_{\text{open}}$ has $\sum_M 1/d \leq U$, but covering needs $\sum_M 1/d \geq 1$ (Prop. 2.4). Eliminates $> 99.7\%$ of modulus subsets: counts 67/32768 (945), 431/131072 (1575), 85/131072 (2205), 1709/524288 (2835). Minimal k achieving threshold 10 (945, 1575, 2835) and 12 (2205), not 15.

2. **Uncovered-residue density deficit (SAFE)**. Maintain bitset of covered residues $U_{\text{cov}} = \bigcup_{d \in M_{\text{fixed}}} (a_d + d\mathbb{Z}) \cap [0, L-1]$, $u = |U_{\text{cov}}|$. Each open d contributes at most L/d new residues (optimistic disjointness), so

$$u + \sum_{M_{\text{open}}} L/d < L \Rightarrow \text{prune.}$$

Stronger than reciprocal when overlaps among fixed d large. Maintained incrementally bitset size $L \leq 2835$ trivial.

3. **Divisibility / LCM anchoring (SAFE)**. From Thm 2.3, any odd covering's L needs $9 \mid L$ or $15 \mid L$. So for fixed L with $9 \mid L$ or $15 \mid L$, feasible M must contain a multiple of 9 or 15: clause $\bigvee_{d: 9 \mid d \vee 15 \mid d} y_d$ (for 945: $y_9 \vee y_{45} \vee y_{135} \vee \dots$). Also optional $\text{lcm} = L$ cut $\forall p^e \parallel L \bigvee_{d: p^e \mid d} y_d$. Sound.

4. **CRT interference (SAFE for counting, HEURISTIC for branching)**. For fixed d_1, d_2 with residues a_1, a_2 , let $g = \gcd(d_1, d_2)$,

$$|(a_1 + d_1\mathbb{Z}) \cap (a_2 + d_2\mathbb{Z}) \cap [0, L-1]| = \begin{cases} L/\text{lcm}(d_1, d_2) & a_1 \equiv a_2 \pmod{g}, \\ 0 & \text{otherwise (incompatible)}. \end{cases}$$

Incompatible pairs are disjoint (good for covering), compatible overlap in L/lcm . For upper-bounding future coverage we assume best-case incompatibility among open moduli (overestimates coverage); if even optimistic bound $< L$, prune is safe. Unit propagation: if n uncovered and only one open d can cover n , then $x_{d, n \bmod d}$ forced.

5. **Forced residue / incompatibility elimination (SAFE)**. Same CRT principle propagates forced assignments, not discard.

6. **Symmetry normalization: translation invariance (SAFE).** Covering constraints invariant under C_L action $x_{d,a} \mapsto x_{d,(a+t) \bmod d}$, $t \in \mathbb{Z}/L\mathbb{Z}$ (adding t to all a_i preserves covering). Every orbit contains translate with $a_3 = 0$. Impose $x_{3,0} = 1$, $y_3 = 1$ as symmetry break (factor 3 reduction; with lex-leader up to L -fold). Sound because we explore canonical representative only.
7. **Lower bound on remaining capacity (SAFE, subsumed by 2).** Same as deficit.
8. **Branching order (HEURISTIC).** Prefer next d with large L/d and small gcd with fixed M (maximises potential new coverage) or small d first; choose residue a that maximises incompatibility with fixed. This is search ordering, not pruning: does not imply unsatisfiability, only affects solver speed. Must try all $a \in [0, d-1]$ in completeness.

We clearly distinguish: 1,2,3,4 (optimistic counting),6 are **provably safe** (prune only unsatisfiable branches). 8 and tie-breakers are **heuristic ordering**.

9. Computational Results

Executed `ANSWER/verify_solution.py` (exact rational arithmetic, SymPy). Results:

L	V	pairwise AMO	seq AMO	cover clauses	$ D $
945	1919	532040	5757	945	15
1575	3223	1479413	9669	1575	17
2205	4445	2897310	13335	2205	17
2835	5807	4794746	17421	2835	19

ILP vars $V + |D| = 1934, 3240, 4462, 5826$; constraints $|D| + L = 960, 1592, 2222, 2854$; literals $L \cdot |D| = 14175, 26775, 37485, 53865$; naive assignments $\prod d \log_{10} 23.80\text{--}34.53$.

Subset pruning: 67, 431, 85, 1709 supports survive reciprocal filter out of 32768, 131072, 131072, 524288 (0.20%, 0.33%, 0.065%, 0.33%). Combined optimism leaves $\sim 10^{13}\text{--}10^{15}$ leaf residue nodes – intractable for exhaustive enumeration without CDCL learning; SAT portfolio with cube-and-conquer on y_d cubes (67 cubes for 945) is at edge of feasibility: $L = 945$ **plausibly solvable in hours–days on multi-core cluster (sequential AMO, y first), larger L expected infeasible without sieve global cut**. No SAT solver was run to completion here; we certify encoding dimensions and pruning power, not a SAT UNSAT certificate. Distinguishing: “search did not find cover” $\neq$ “proved no cover”; per L need complete UNSAT proof (DRAT) for certified non-existence conditional on L .

Verification of classic even covering $\{(0, 2), (0, 3), (1, 4), (1, 6), (11, 12)\}$ covers $\mathbb{Z}/12\mathbb{Z}$ (12/12). Small odd trial $\{(0, 3), (1, 5), (2, 7)\}$ covers only 57/105 residues, correctly fails.

10. Independent Verification

- **SymPy exact:** divisor tables, $\sigma(L)$, $\tau(L)$, $\sum_{d|L, d>1} 1/d$, $\sigma(L)/L$, abundant equivalence $\sigma(L)/L = 1 + \sum_{d|L, d>1} 1/d$, $L \geq 945$ minimality, and subset counts recomputed independently in script and printed; all residuals 0 (exact Rational equality). Command: `python3 ANSWER/verify_solution.py 2>&1 | tail -n 80` – exit code 0, `assert` passed for folklore identity, classic covering 12/12, and odd $L < 945$ exhaustive 0 successes.
- **Literature live fetch:** <https://www.erdosproblems.com/7> (title, VERIFIABLE Open, \$25, tags, citations [BBMST22] [HoNi19] [Sc67] [FFK00], last edited 2026-01-22), /`latex/7`, /`forum/thread/7` 21 comments including #6183 monotonicity claim, #6298 refutation, #6316 Bloom block, arXiv:1811.03547 and arXiv:1703.02133 abstracts – all match cited theorems (see §2).

- **Numerical residuals:** $R := \sum_{d|L, d>1} 1/d - (\sigma(L)/L - 1) = 0$ exactly for $L \in \{945, 1575, 2205, 2835, 3465, \text{(printed 0)}\}$. Pairwise clause counts cross-checked $\sum \binom{d}{2}$ exact.
- **Lean:** No formal Lean proof attempted here beyond sketch in §11; previous Lean attempts refuted (false `sieveProd<1`). We do not claim Lean compilation; verification is CAS/numerical.
- **Distinction proof vs evidence:** SAT size feasibility argument is complexity estimate, not proof of UNSAT; noted as heuristic unless solver emits DRAT certificate. Sieve extension remains conditional (see §11).

11 11. Gap Analysis

11.1 Critical self-audit (10 checks)

1. *Did I assume every divisor of L may occur as modulus?* No. We treat $M \subseteq D(L)$ variable; covering condition uses subset. Reciprocal inequality $\sum_{d|L, d>1} 1/d \geq \sum_M 1/d$ is correctly subset direction; we never reversed. Explicitly distinguished M vs $D(L)$.
2. *Did I replace inequality over selected moduli with inequality over all divisors?* Checked. Prop. 2.4 states $\sum_{D(L)} 1/d \geq \sum_M 1/d \geq 1$ is necessary; sufficiency not claimed. Pruning invariant 1 uses U over $M_{\text{fixed}} \cup M_{\text{open}} \supseteq M$, sound upper bound.
3. *Did I assume squarefreeness where not available?* No. Prime-power section explicitly isolates where SF used and shows PP requires different weights $\sum p^{-j}$ and larger $c_0^{(\gamma)}$. We do not claim SF theorem implies PP.
4. *Did I assume independence between prime-power fibres?* No. Obstruction Theorem shows fibres with $\gamma > 1$ couple via mixed p^{-j} weights and E_i cross terms; variance inflates $2.09\times$ at $p = 3, \gamma = 3$; we state independence estimate fails without new bound.
5. *Did I assume candidate L set exhaustive?* No. §7 proves $\{945, 1575, 2205, 2835\}$ are four smallest with $9 \mid L$ or $15 \mid L$ up to 4000 and proven minimal 945, but explicitly flag that larger L (e.g. 3465) not multiples of members, so finite search does not imply global non-existence; sieve needed.
6. *Did I confuse failed search with non-existence?* No. §9 distinguishes `SEARCH_DID_NOT_FIND` vs `PROVED_NO_COVER`; we report pruning power and feasibility estimate, not UNSAT certificate; we state $L = 945$ plausibly solvable but not solved here.
7. *Did I convert numerical approximation into theorem?* No. Values $c_0(1) \approx 0.098$, $c_N(1) \approx 0.612$ cited as BBMST numerical evaluation for $N = 500$ with explicit δ table, not proved exact here; we treat as computed datum, note that PP analogue requires recomputation, and do not claim $c_N^{(\gamma)} < 1$ without LP certificate.
8. *Did I introduce new density-update formula without derivation?* No. Operator $(\star)$ derived from BBMST (16),(23),(34)–(36),(47); we reproduce derivation steps and highlight where naive formula $1 + x/((1 - \delta)(p^\gamma - 1))$ is false and correct prefactor $(\sum p^{-j})^2$ derived.
9. *Did I cite secondary summary instead of actual theorem?* No. Cite primary Invent. Math. 230 and Duke 168 papers plus arXiv 1811.03547/1703.02133 and MR numbers; quote p. 8 verbatim on SF requiring separate paper.

10. Does proposed conditional lemma genuinely imply original conjecture? Yes. Conditional Lemma C (certified LP $c_0^{\text{pow}}(1) \leq c_0^{\text{SF}}(1)$ plus feasible $\delta^{(\gamma)}$) yields $c_N^{(\gamma)}(1) < 1 \Rightarrow \mu(\mathbb{Z} \setminus \bigcup A_d) > 0 \Rightarrow$ no covering, via *existing* BBMST induction Sections 4–5 unchanged; we state exact lemma and why PP tail monotonicity alone insufficient but lemma sufficient.

11.2 Strongest conditional lemma that would bridge gap

Lemma 11.1 (Conditional Lemma C, LP verification, quantitative, falsifiable). *Let $S_0^{\text{pow}} = \{m \leq 73 : m \text{ odd}\}$ (27 numbers) vs S_0^{SF} (16). Let $\mathcal{F}$ be BBMST LP feasible weights over S_0^{pow} (fibre capacities). Then optimum $c_0^{\text{pow}} := \max_{\mathcal{F}} \sum w_m f_m(\delta)$ satisfies $c_0^{\text{pow}} \leq 0.098$ (SF optimum). This is finite LP with $\sim 10^4$ constraints, verifiable by exact rational/interval arithmetic. Combined with tail bound $\prod_{k \geq 6} (1 + \eta_k^{(\gamma)} / (1 - \delta_k^{(\gamma)})) \leq \prod_{k \geq 6} (1 + 1 / ((1 - \delta_k^{\text{SF}})(p_k - 1)))^{(\text{reopt})}$ after exponent-aware re-optimisation of $\delta^{(\gamma)}$, gives $c_N^{(\gamma)}(1) < 1$.*

If Lemma 11.1 proved, then BBMST distortion measure gives positive uncovered density uniformly for any finite odd distinct family, hence **no odd distinct covering exists** (full Erdős–Selfridge). Failure of lemma would certificate that distortion method as calibrated cannot extend without new δ shape or larger N / case split $9 \mid L$ vs $15 \mid L$.

Barrier classification: **METHOD REACHES IDENTIFIED TECHNICAL BARRIER** – initial-segment LP and fibre combinatorics for p^e trees, not intrinsic impossibility of sieve philosophy. Heuristic suggests extension plausible but must be computed.

12 12. Final Conclusion

Erdős Problem 7 is UNRESOLVED (OPEN, verifiable).

- **Exactly what is known (proved):**

1. No squarefree odd distinct covering (BBMST22, $c_N(1) \approx 0.612 < 1$).
2. Any distinct covering has $2 \mid d$ or $3 \mid d$ (HoNi19, simplified BBMST).
3. Any odd distinct covering has L with $9 \mid L$ or $15 \mid L$ (BBMST22 refinement).
4. Folklore $\sum_{d \mid L, d > 1} 1/d \geq 1 \iff \sigma(L) \geq 2L$, so odd $L \geq 945$ sharp; no odd $L < 945$ passes (exhaustive, residual 0).
5. Two 2026 axiomatic Lean attempts invalid (false `sieveProd<1`, omitted c_0).

- **Exactly what this investigation proves:**

1. Reconstruction of BBMST operator $c_N(x) = c_0(x) \prod (1 + x / ((1 - \delta_k)s_k))$ in actual notation with $c_0(1) \approx 0.098$, $s_k = p_k - 1$, $\delta_k \in (0, 1/2)$, moment bounds (36)(47), ν -product, and Θ -recursion.
2. Rigorous prime-power obstruction Theorem (first obstruction at α_i fibre definition): prefactor increases with γ , LP feasible region larger, δ exponent-dependent, naive $p^\gamma - 1$ monotonicity sign-reversed.
3. Exact finite SAT/ILP encoding per L with dimensions $V = 1919\text{--}5807$, $\prod d \cdot 10^{23.80\text{--}34.53}$, covering clauses L , AMO $\sum \binom{d}{2}$ or $3V$.
4. Candidate L exact tables (945, 1575, 2205, 2835, 3465), divisor sets, reciprocal sums 1.016–1.049, minimal $k = 10\text{--}12$, headroom < 0.05 , not assumed exhaustive.
5. Provably safe pruning invariants (reciprocal, deficit, LCM anchoring, CRT optimistic, translation $a_3 = 0$) vs heuristic ordering, quantified elimination 99.67%–99.93% (67,431,85,1709 survivors).

- **Exactly what it does NOT prove:**

1. No proof of full Erdős–Selfridge conjecture (allowing prime powers).
 2. No global exhaustive enumeration beyond $\mathcal{L}$; no SAT UNSAT certificate for any L produced here.
 3. No certification of Lemma 11.1 $c_0^{\text{pow}} \leq c_0^{\text{SF}}$; requires recomputing LP over 27 numbers and re-deriving Θ recursion for p^e branching.
- **Precise analytical obstruction:** Fibre coupling $\alpha_i^{(\gamma)} = \sum_j p^{-j} \mathbf{1}[\dots] + E_i$ makes second moment $\propto (\sum p^{-j})^2 = (1 - p^{-\gamma})^2 / (p - 1)^2$ increase with γ , not decrease as $1/(p^\gamma - 1)^2$. At $p = 3, \gamma = 3$ this is $2.09\times$ larger than SF, reversing naive heuristic $169\times$ reduction. First step where squarefreeness essential.
 - **Technical vs intrinsic:** Technical gap in $p \leq 73$ LP and p^e -tree bad-fibre bounds. No evidence of intrinsic wall: distortion method flexible, prime powers enlarge block sizes intuitively helping sieve, but LP optimum may increase; outcome depends on finite computation not yet performed. BBMST authors deliberately left SF as separate paper, not trivial corollary.
 - **Strongest conditional lemma:** Lemma 11.1 (certified LP over S_0^{pow} with re-optimised $\delta^{(\gamma)} \Rightarrow c_N^{(\gamma)}(1) < 1 \Rightarrow$ positive uncovered density $\Rightarrow$ impossibility. Equivalent measure-theoretic and distortion-bound formulations given in §11.
 - **What follows if proved:** Full resolution of Erdős Problem 7 negative (no odd distinct covering), unifying squarefree and general odd cases under single distortion framework; plus axiom-free Lean formalisation becomes feasible modulo LP certificate.
 - **What computational searches actually ruled out:** Level: dimensions and pruning power certified; no L in $\mathcal{L}$ ruled out or exhibited here beyond necessary condition $\sum 1/d \geq 1$ (which all four pass). Search scale 10^{23} – 10^{34} shows naive enumeration impossible; $L = 945$ at edge of portfolio SAT with cube-and-conquer (67 cubes), larger L infeasible without sieve induction. Any claim “no cover up to $L = 2835$ ” would require DRAT proof not produced.

Synthesis assessment: METHOD REACHES AN IDENTIFIED BARRIER / PARTIALLY RESOLVED / UNRESOLVED – squarefree case closed, general odd case blocked at prime-power LP/distortion coupling; conditional lemma precisely isolates gap. Until Lemma 11.1 computed, Erdős–Selfridge remains heuristic (0.612 margin suggests true) but open, verifiable by finite example (§25) if counterexample found.

Artifacts & Verification

- ANSWER/report.tex (this file) and ANSWER/report.pdf compiled via pdflatex.
- ANSWER/verify_solution.py – prints candidate L , factorisations, divisor counts, V , $\log_{10} \prod d$, reciprocal sums, σ , abundant checks, exhaustive $L < 945$ scan, subset counts after $\sum 1/d \geq 1$ filter, CNF/ILP dimensions, classic covering 12/12, small odd failure 57/105, BBMST mock $c_0 \cdot \prod$ vs naive sieveProd > 1. Residuals 0 exactly.
- Live sources fetched 2026-08-25: <https://www.erdosproblems.com/7,/latex/7,/forum/thread/7,github.com/teorth/erdosproblems,arXiv:1811.03547,arXiv:1703.02133>.

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